

Durum	Biten
Başladı	Cuma, 18 Nisan 2025, 2:30 PM
Tamamlandı	Cuma, 18 Nisan 2025, 4:00 PM
Süre	1 saat 29 dk
Not	100,00 üzerinden 61,00

Soru bilgi

Note: This must be written in your own words. Anything copied from the slides/text will be considered incorrect.
Explain the following terms

Soru 1
Tamamlandı
5,00 üzerinden 5,00 notunu ver

Functional Dependency

A constraint between two attributes or sets of attributes where the value of another attributes. Denoted as $A \rightarrow B$, meaning B is functionally dependet on A.

Yorum:

Soru 2
Tamamlandı
5,00 üzerinden 5,00 notunu ver

Partial Dependency

A partial dependency happens when a no-prime attribute is dependent on part of a composite primary key, not the whole key.for example, If table has a composite key (A, B), and C depends only on A, then C is partially dependent on the key.

...
It leads to unnecessary data duplication. Removed in 2nf. Thats why it is a problem

Yorum:

Soru 3

Tamamlandı

5,00 üzerinden 5,00 notunu ver

Transitive Dependency

A transitive dependency occurs when a non-prime attribute depends on another non-prime attribute indirectly through a second non-prime attribute. for ex. if $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$ is a transitive dependency. so it can cause redundancy and update anomalies. Eliminated in 3nf. thats why it is a a problem.

Yorum:

Soru 4

Tamamlandı

5,00 üzerinden 5,00 notunu ver

Full Functional Dependency

An attribute is fully functional dependent on a set of attributes. If it is functionally dependent on the entire set but not on any proper subset of it.

Yorum:

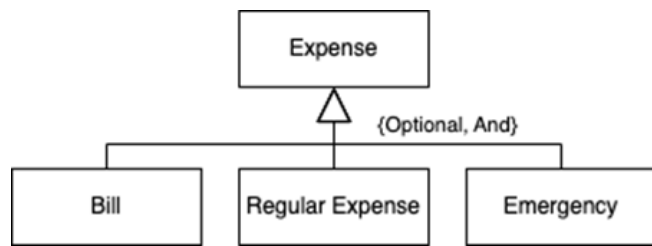
Soru bilgi

As part of a budgeting application, your team is tasked with classifying Expenses. Two members of your team each submit a portion of the ER-model to you. What are the differences between how each member understands how the entities relate to each other? Write your response in the box below each diagram.

Soru 5

Tamamlandı

5,00 üzerinden 5,00 notunu ver



An expense can optionally be any combination of Bill, Regular Expense and Emergency. This means a single Expense might be none, zero, one, two or all three of subtypes.

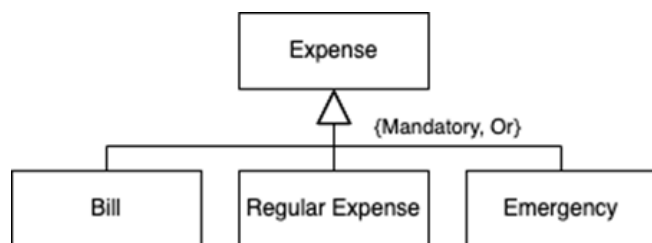
For example, an expense could be just a bill, or both a regular expense and an emergency, or even all three at once.

Yorum:

Soru 6

Tamamlandı

5,00 üzerinden 5,00 notunu ver



An expense must be exactly one of the following: Bill, Regular Expense, or Emergency.

This enforces that every Expense must belong to one and only one subtype.

For example, an expense must either be a bill or a regular expense or an emergency, not a combination..

Yorum:

Soru 7

Tamamlandı

Not verilmemiş

Only complete this question if the room you should have been in was full.

Which room should you have been in for the room bonus code?

- ☐ a. Yazilim Lab 1
- ☐ b. Yazilim Lab 2
- ☒ c. Yazilim Lab 3

Cevabınız doğru.

Doğru cevaplar: Yazilim Lab 1, Yazilim Lab 2, Yazilim Lab 3

Soru bilgi

Mission Statement: We want to develop a database for managing our home monthly Budget. Some example data is given below. Note: This example data is only from the most recent months but should not be assumed to be the only data in our source data. You may use the letter column headers for questions where those would be appropriate responses.

A	B	C	D	E	F	G	H	I	J
Year	Month	Category	Item	Payment Method	Recipient Type	Recipient	Date Due	Estimated	Actual
2025	March	Housing	Rent	Bank Transfer	Person	Ev Sahibi	1-Mar-2025	9000	9000
		Utilities	Mobile Phone	Bank Transfer	Communications Company	Turkcell	7-Mar-2025	450	450
			Electricity	BillPay	Utility Company	CK Akdeniz	15-Mar-2025	2000	2062.5
			Water	BillPay	Utility Company	ASAT	21-Mar-2025	400	385.23
			Internet	BillPay	Communications Company	Turkcell	25-Mar-2025	580	580
		Food	Groceries	Cash	Grocers	Markets		2500	2625
			Dining out	Credit Card	Food Service	Restaurants and Cafes		1000	980
		Education	Books	Credit Card	E-Commerce	Amazon		800	800
	April	Housing	Rent	Bank Transfer	Person	Ev Sahibi	1-Apr-2025	9000	9000
		Utilities	Mobile Phone	BillPay	Communications Company	Turkcell	6-Apr-2025	450	450
			Electricity	BillPay	Utility Company	CK Akdeniz	16-Apr-2025	2000	1812.13
			Water	BillPay	Utility Company	ASAT	23-Apr-2025	400	
			Internet	BillPay	Communications Company	Turkcell	24-Apr-2025	580	
		Food	Groceries	Cash	Grocers	Markets		2700	
			Dining out	Credit Card	Food Service	Restaurants and Cafes		1200	

Soru 8

Yanlış

2,00 üzerinden 0,00 notunu ver

The example data is in which of the following?

- ☐ a. UNF
- ☒ b. 1NF 
- ☐ c. 2NF
- ☐ d. 3NF
- ☐ e. Other

Cevabınız yanlış.

Doğru cevap: UNF

Soru 9

Tamamlandı

8,00 üzerinden 4,00 notunu ver

Explain how you know the data is in the form you selected in the previous question.

I think the data is in 1NF because each field contains just one value. Each row is unique, with no duplicates, all of them contain atomic values.

Yorum:

Soru 10

Kismen doğru

5,00 üzerinden 4,00 notunu ver

Identify a Candidate/Primary Key for the Budget relation

Cevap: Year, Month, Category, Item



Doğru cevap: A, B, D

Yorum:

Soru 11

Tamamlandı

10,00 üzerinden 6,00 notunu ver

Identify functional dependencies that exist. Note: all columns should be used at least once.

Year, Month, Category, Item, Date Due > Payment Method, Recipient Type, Recipient, Estimated, Actual . Because the rest can be determined. Recipient > Recipient type, recipient name. if we know recipient, we can determine the types (person, communications, company) and their name.

Yorum:

Soru 12

Yanlış

5,00 üzerinden 0,00 notunu ver

Identify any partial dependencies that exist.

Cevap: Partial dependencies exist when no-prime attributes are dependent on a part of a prime attribute. 

Doğru cevap: D -> C, E, F, G

Soru 13

Kismen doğru

5,00 üzerinden 2,00 notunu ver

Identify any transitive dependencies that exist.

Cevap: Recipient -> Recipient Type 

Doğru cevap: D -> F, G and G -> F

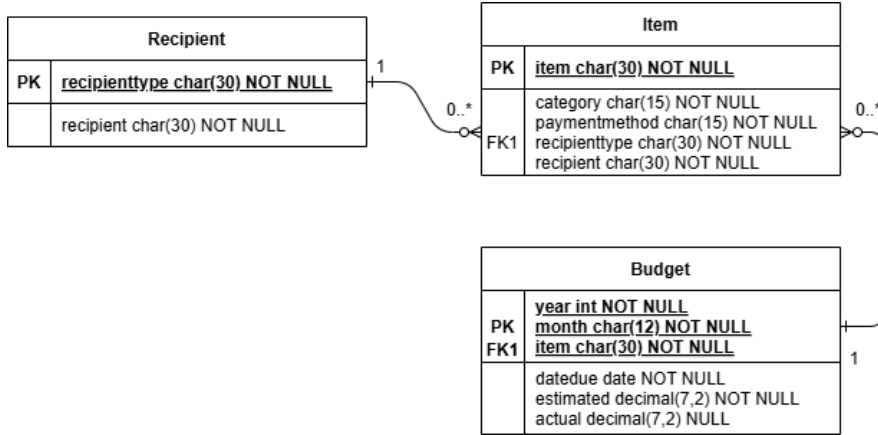
Yorum:

Soru 14

Tamamlandı

20,00 üzerinden 3,00 notunu ver

The following ER model was developed to be a normalized representation of the Budget database. Identify three (3) errors in this design.



1- Unnecessary redundancy in the item table:

- The item table has both recipient and recipienttype as part of its composite primary key. A single item can belong one or more recipients but these fields not need to be part of the primary key of the item. May be we can remove the recipient and recipienttype from the Item table and store them in a separate table that links items and recipients. Maybe ItemRecipient

2- The budget table lacks recipients info.

The budget table references item but doesnt contain any information about the recipient, such as recipient and recipienttype . Since these are crucial pieces of information to understand who is associated with each budget item, they should be part of the budget table. So I think the budget table should reference both item and recipient information to complete the relationship. We can add recipient and recipienttype as foreign keys in the budget table.

3- missing foreign key constraints for item table.

Yorum:

Soru bilgi

Using the corrected normalized database (after you corrected the above diagram), write SQL queries for each of the following tasks.

NOTE: Each of these must be a single SQL query.

Soru 15

Tamamlandı

3,00 üzerinden 3,00 notunu ver

Give a list of budget items and the total amount budgeted on each of them in 2021.

```
SELECT i.item, (SELECT SUM(b.estimated)
FROM Budget b
WHERE b.item = i.item
AND b.year = 2021)
AS total_budgeted
FROM item i;
```

Yorum:

Soru 16

Tamamlandı

3,00 üzerinden 2,00 notunu ver

Give a list of budget items that have amounts at least 3800

```
SELECT i.item, b.estimated
FROM Budget b
JOIN Item i
ON b.item = i.item
WHERE b.estimated >= 3800
AND b.item
IN (SELECT item FROM Budget
WHERE estimated >= 3800);
```

Yorum:

Soru 17

Tamamlandı

3,00 üzerinden 2,00 notunu ver

Remove all budget items that are older than 2 years old (before 2023 is enough).

DELETE FROM Budget WHERE year < 2023 AND item IN

Yorum:

Soru 18

Tamamlandı

3,00 üzerinden 3,00 notunu ver

We want to start our budget plan for May 2025. Add all the same information from April, but make the Month "May". Do not remove any rows for April! You should copy the rows from April to May.

```
INSERT INTO Budget (year, month, item, datedue, estimated, actual)
SELECT 2025, 'May', item, datedue, estimated, actual
FROM Budget WHERE year = 2025 AND month = 'April';
```

Yorum:

Soru 19

Tamamlandı

3,00 üzerinden 2,00 notunu ver

Change the value for rent for April to be %25 more than what it was before.

```
UPDATE Budget SET estimated = (  
    SELECT estimated * 1.25 FROM Budget  
    WHERE item = 'Rent' AND month = 'April' AND year = 2025  
) ,  
actual = (  
    SELECT actual * 1.25 FROM Budget  
    WHERE item = 'Rent' AND month = 'April' AND year = 2025  
    ) WHERE item = 'Rent' AND month = 'April' AND year = 2025;
```

Yorum: